higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 2

(12041032)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 The earth leakage test button is used to check ...

A continuity.
B that the trip mechanism operates correctly.
C the sensitivity of the earth leakage unit.
D polarity.

1.1.2 The voltage drop from the point of supply to the point of consumption must not be more than:

A 1%
B 10%
C 5%
D 15%

1.1.3 Disconnect and inform the supplier if the voltage exceeds … while doing an elevated voltage on neutral test.

A 25 V
B 10 V
C 15 V
D 50 V

1.1.4 When doing an insulation resistance test, ensure that the test voltage is at least … the supply voltage.

A the same as
B half
C four times
D twice

1.1.5 A/an … test is done to ensure that the consumer's earth terminal and all earthing points in the installation are connected together.

A earth resistance
B insulation resistance
C resistance of earth continuity conductor
D earth loop impedance

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1.1.6 The supply must be ON, when conducting a/an …

A insulation resistance test.
B continuity of bonding test.
C earth continuity test.
D operation of earth leakage test.

1.1.7 When conducting an insulation resistance test, an acceptable reading will be:

A 1 000 Ω
B 10 000 Ω
C 100 000Ω
D 1 000 000 Ω

1.1.8 A multimeter set on the lowest ohm scale, can be used to conduct the following test:

A Resistance of the earth continuity conductor
B Earth loop impedance
C Earth resistance
D Insulation resistance

1.1.9 Conductive roofs shall be bonded and earthed and the resistance of the earth continuity path shall not exceed:

A 0,01 Ω
B 0,2 Ω
C 0,5 Ω
D 0,9 Ω

1.1.10 To issue a certificate of compliance, the person must …

A be studying electrical engineering.
B have done a house wiring course.
C be accredited.
D have completed a trade test.

(10 × 1) (10)

1.2 Draw a neat, fully labelled circuit diagram of a single-pole switch controlling ONE light. The diagram should include a distribution board showing the components required for the installation. (10)

1.3 List FIVE signs of poor housekeeping in the workshop. (5)

[25]

QUESTION 2

2.1 Refer to the FIGURE 1 below, and answer the questions.

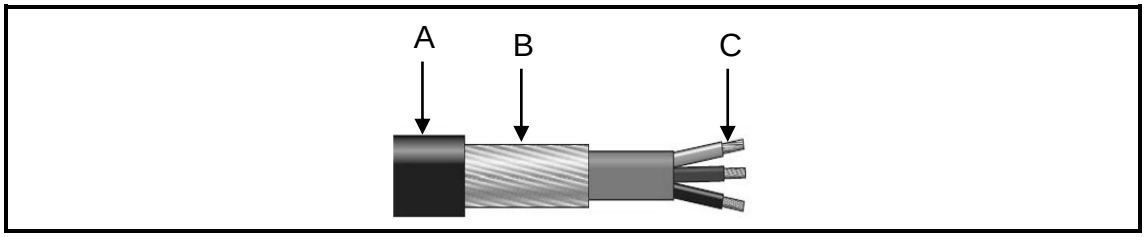


FIGURE 1

2.1.1 Identify the cable in FIGURE 1. (1)

2.1.2 Identify the parts of the cable. Write only your answer next to the letter (A–C) in the ANSWER BOOK. (3)

2.1.3 What is the purpose of part B? (2)

2.2 The cable in FIGURE 1 needs to be joined after it was damaged.

2.2.1 What joining kit would you use? (1)

2.2.2 Briefly explain how you would join the cable. (6)

2.3 After the joint was completed, a continuity test was done and an infinity reading was recorded on one of the conductors. (2)

Is this reading acceptable? Substantiate your answer.

2.4 Trunking allows many cables/conductors to be installed in an electrical installation.

List FIVE advantages of using trunking over other methods of housing cables. (5)
[20]

QUESTION 3

3.1 Motors must be tested regularly to ensure that they are in good condition.

Name THREE electrical tests that can be conducted on a motor. (3)

3.2 A capacitor-start capacitor-run motor fails to start. You are required to find the fault.

Name FOUR components in the motor that you can test while doing fault-finding. (4)

3.3 A DC compound motor has TWO field coils, a series and shunt coil.

Show by means of a neat, labelled diagram how you would connect these coils as a long-shunt compound motor. (5)

3.4 The main function of a motor control is to control the operation of motors.

List THREE operational functions of a motor control. (3)

[15]

QUESTION 4

4.1 Complete the table below. Write only your answer next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

Test on component	Measuring instrument	Acceptable value
Sensitivity of an earth leakage relay	4.1.1	4.1.2
Continuity of a fuse (with power off)	4.1.3	4.1.4
Insulation resistance test on a transformer	4.1.5	4.1.6
Continuity of a switch (closed position)	4.1.7	4.1.8
Insulation resistance test on a cable	4.1.9	4.1.10

(10 × 1) (10)

4.2 Polarity tests must be done on an electrical installation.

Give a reason for conducting a polarity test on a lamp holder. (3)

4.3 While doing a continuity test on an isolator in the off position, a 0 Ω reading was recorded.

Is this reading acceptable? Substantiate your answer. (2)

[15]

QUESTION 5

5.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (5.1.1–5.1.10) in the ANSWER BOOK.

5.1.1 After a meter has been replaced, it must be tested to ensure it works correctly.

5.1.2 When using a digital meter, always be aware of parallax errors.

5.1.3 A wattmeter only consists of a current coil connected in series with the load.

5.1.4 A frequency meter is used to measure the power factor of an AC electric power system.

5.1.5 Load control relays are used to shed certain loads during peak periods.

5.1.6 Discard all waste materials wherever you want to on completion of the task.

5.1.7 Work schedules only need to be communicated to the person installing the new metering unit.

5.1.8 Planning properly will help avoid making costly mistakes.

5.1.9 It is not necessary for an electrician to acquire authorisation into a restricted area as he is qualified and knows what he is doing.

5.1.10 Equipment and tools must be regularly inspected to ensure they are in good working order and safe to use.

(10 × 1) (10)

5.2 With the aid of a neat, labelled diagram, show how you would connect a voltmeter with a potential transformer to measure a voltage of 11 kV. (5)

[15]

QUESTION 6

6.1 You are required to install TWO 12 V-0,5 A, PV modules for a client at his home.

6.1.1 With the aid of a neat diagram, show how you would connect the modules in parallel. (2)

6.1.2 Determine the total voltage of the combination. (1)

6.1.3 Determine the total current of the combination. (2)

6.2 Complete the following paragraph by choosing a word(s) from the list below. Write only the word(s) next to the question number (6.2.1–6.2.5) in the ANSWER BOOK.

reflects; electricity; holes; negatively; photons; protons; free; shines; positively

When sunlight (6.2.1 …) on the surface of a PV module, particles of light called (6.2.2 …) bounce into (6.2.3 …) charged electrons to become (6.2.4 …). This movement of electrons generates (6.2.5 …). (5)

[10]

TOTAL: 100

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